HOCHSCHULE OSNABRÜCK
UNIVERSITY OF APPLIED SCIENCES

WORKING WITH THE HPC CLUSTER

hard facts

compute CPUs:	1.536
compute RAM:	15.360 GB
GPUs:	32 x A100 (80 GB) 14 x A100 (40 GB)
storage:	672 TB (HDD) 207 TB (SSD)

hard facts

compute CPUs:	1.536
compute RAM:	15.360 GB
GPUs:	32 x A100 (80 GB) 14 x A100 (40 GB)
storage:	672 TB (HDD) 207 TB (SSD)

enables powerful machine learning applications (inference and training)

https://docs.hpc.hs-osnabrueck.de

https://web.hpc.hs-osnabrueck.de

access from within the university's network
using a modern web browser

Required steps to get started:

1. open SSH shell
2. copy virtual environment
3. start new JupyterLab instance
4. clone provided git repository

Required steps to get started:

1. open SSH shell
2. copy virtual environment
3. start new JupyterLab instance
4. clone provided git repository

HiPer4All | Apps | Files | Jobs | Interactive Apps | My Interactive Sessions | Help | Logged in as nimeseth | Log Out

- HiPer4All Desktop
- Jupyter Lab
- HiPer4All Shell Access

HOCHS
UNIVERSIT

Welcome to the HiPer4All Dashboard. From here you can access the HPC Cluster resources through your Web Browser!

Pinned Apps A featured subset of all available apps

Jupyter Lab
System Installed App

HiPer4All Desktop
System Installed App

Fair Use Policy

```
Welcome to the HiPer4All HPC Cluster!
By using our services, you agree to the
Cluster's Fair Use Policy!
You can read it at https://docs.hpc.hs-
osnabrueck.de/en/pages/overview/fair-use.html
```

Host: hpc.hs-osnabrueck.de

Themes: Default

```
Welcome to the HiPer4All HPC Cluster!
By using our services, you agree to the Cluster's Fair Use Policy!
You can read it at https://docs.hpc.hs-osnabrueck.de/en/pages/overview/fair-use.html

Last login: Tue May 21 14:30:47 2024 from 172.25.198.21
Linux 4.18.0-513.24.1.el8_9.x86_64 - Rocky Linux release 8.9 (Green Obsidian)
[nimeseth@m10-09 ~]$
```

Required steps to get started:

1. open SSH shell
2. copy virtual environment
3. start new JupyterLab instance
4. clone provided git repository

Host: hpc.hs-osnabrueck.de

Themes: Default

```
[nimeseth@m10-09 ~]$ cp -r /cluster/groups/l_aul_meseth_bda_ss24/env /cluster/user/nimeseth/env
```

```
cp -r /cluster/groups/l_aul_meseth_bda_ss24/env /cluster/user/<user>/env
```

Replace with your username

Required steps to get started:

1. open SSH shell
2. copy virtual environment
3. start new JupyterLab instance
4. clone provided git repository

HiPer4All Apps Files Jobs Interactive Apps My Interactive Sessions Help Logged in as nimeseth Log Out

HOCHSCHULE OSNABRÜCK
UNIVERSITY OF APPLIED SCIENCES

Welcome to the HiPer4All Dashboard. From here you can access the HPC Cluster resources through your Web Browser!

Pinned Apps A featured subset of all available apps

Jupyter Lab
System Installed App

HiPer4All Desktop
System Installed App

Fair Use Policy

```
Welcome to the HiPer4All HPC Cluster!
By using our services, you agree to the
Cluster's Fair Use Policy!
You can read it at https://docs.hpc.hs-
osnabrueck.de/en/pages/overview/fair-use.html
```

HiPer4All Apps Files Jobs Interactive Apps My Interactive Sessions Help Logged in as nimeseth Log Out

HOCHSCHULE OSNABRÜCK
UNIVERSITY OF APPLIED SCIENCES

Welcome to the HiPer4All Dashboard. From here you can access the HPC Cluster resources through your Web Browser!

Pinned Apps A featured subset of all available apps

Jupyter Lab
System Installed App

HiPer4All Desktop
System Installed App

Fair Use Policy

```
Welcome to the HiPer4All HPC Cluster!
By using our services, you agree to the
Cluster's Fair Use Policy!
You can read it at https://docs.hpc.hs-
osnabrueck.de/en/pages/overview/fair-use.html
```

HiPer4All Apps Files Jobs Interactive Apps My Interactive Sessions Help Logged in as nimeseth Log Out

Home / My Interactive Sessions / Jupyter Lab

Interactive Apps

HiPer4All Desktop

Jupyter Lab

Jupyter Lab

This app will launch a Jupyter Lab server on a compute node.

Account

l_aul_meseth_bda_ss24

Select the Account on which to bill the Ressources for this Job on.

Python Virtual Environment

/cluster/user/nimeseth/env

You need to select a virtual environment to run your notebook server. This allows you to install custom packages for your programs. You can also create multiple environments for different projects or tasks. See the documentation on how to create a new virtual environment and install Jupyter.

Extra Modules

FFmpeg git

Define any custom modules you want to use inside your notebook server, which are available via lmod and might have been installed via EasyBuild. The input has to be a space-separated list of modules installed in /opt/ohpc/pub/easybuild/modules/all/

Max Time (in h)

8

Must match the path where you copied the virtual environment in the previous step

Load required modules, separated by a space

HiPer4All Apps Files Jobs Interactive Apps My Interactive Sessions Help Logged in as nimeseth Log Out

Session was successfully created. ×

Home / My Interactive Sessions

Interactive Apps

- HiPer4All Desktop
- Jupyter Lab

Jupyter Lab (3787)

Queued

Delete

Created at: 2024-05-21 16:32:39 CEST

Time Requested: 8 hours

Session ID: 7b828d4d-bd1e-4226-a82a-6d4a5dcc9520

Account: l_aul_meseth_bda_ss24

Python Virtual Environment: /cluster/user/nimeseth/env

Max Time (in h): 8

GPU Type: ampere80

Please be patient as your job currently sits in queue. The wait time depends on the number of cores as well as time requested.

powered by OPEN OnDemand

OnDemand version: 3.0.3

HiPer4All Apps Files Jobs Interactive Apps My Interactive Sessions Help Logged in as nimeseth Log Out

Session was successfully created. ×

Home / My Interactive Sessions

Interactive Apps

- HiPer4All Desktop
- Jupyter Lab

Jupyter Lab (3787)

1 node | 4 cores | Running

Delete

Host: m09-05

Created at: 2024-05-21 16:32:39 CEST

Time Remaining: 7 hours and 58 minutes

Session ID: 7b828d4d-bd1e-4226-a82a-6d4a5dcc9520

Account: l_aul_meseth_bda_ss24

Python Virtual Environment: /cluster/user/nimeseth/env

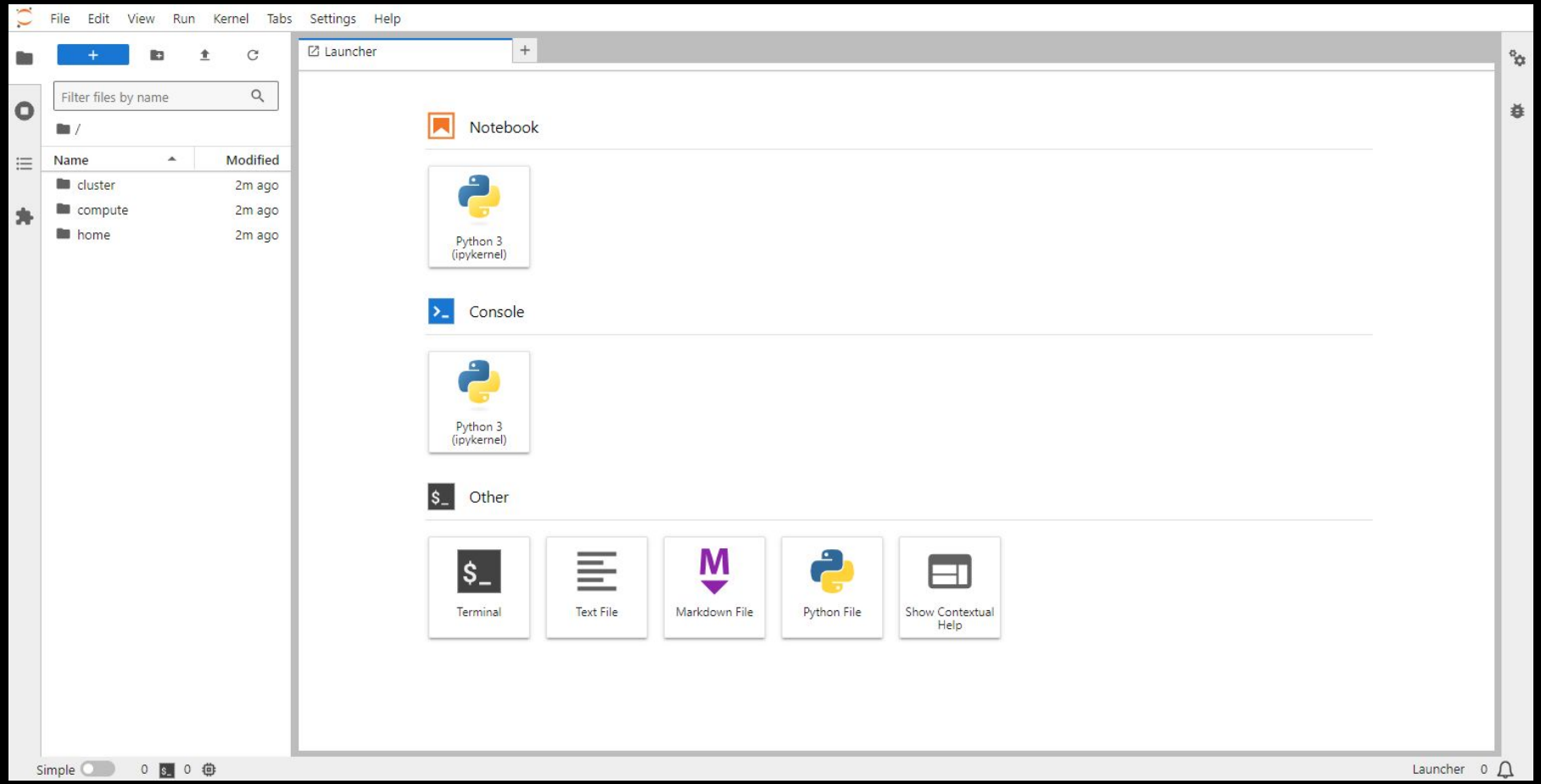
Max Time (in h): 8

GPU Type: ampere80

Connect to Jupyter

powered by

OnDemand version: 3.0.3


File
Edit
View
Run
Kernel
Tabs
Settings
Help
Filter files by name
/
Name
Modified
cluster
2m ago
compute
2m ago
home
2m ago
Launcher
Notebook
Python 3 (ipykernel)
Console
Python 3 (ipykernel)
Other
Terminal
Text File
Markdown File
Python File
Show Contextual Help
Simple
0
0
Launcher
0

Required steps to get started:

1. open SSH shell
2. copy virtual environment
3. start new JupyterLab instance
4. clone provided git repository

Host: hpc.hs-osnabrueck.de

Themes: Default

```
(env) [nimeseth@m10-09 ~]$ pwd
/home/nimeseth
(env) [nimeseth@m10-09 ~]$ git clone https://github.com/winf-hsos/data-analytics-cluster-code.git
```

```
git clone https://github.com/winf-hsos/data-analytics-cluster-code.git
```

useful commands

`ls -a`	List all files and folders, including hidden ones
`rm -r <folder>`	Remove all files and subfolders in <folder>
`cd <folder>`	Change directory to folder
`cd ..`	Move one folder up
`pwd`	Show the current folder
`cp -r <from> <to>`	Copy all content from one folder to another